MATHEMATICS

The Honest Coder

Maths trick #8

Website → https://thehonestcoder-edu.github.io/

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Trick: Multiply two 2-digit numbers

Step by Step Explanation
(with examples)

d1 = first digit of first number
d2 = second digit of first number
d3 = first digit of second number
d4 = second digit of second number

Thus, the first number is (d1 d2) and the second is (d3 d4).

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1. First part: d1 × d3

Multiply both numbers on left.

2. Middle part: (d1 × d4) + (d2 × d3)

Multiply diagonal pairs and add the results.

3. Last part: d2 × d4

Multiply both numbers on right.

Arrange results from left to right.

Then, starting from the right, keep one digit, carry the rest, till the middle and last part has 1 digit. This is the answer.

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Example 1→

23 × 14

Here: d1=2, d2=3, d3=1, d4=4

```
   2 3
 X 1 4
-------
 2 | 8 + 3 | 12
```

First part = 2*1 = 2

Second part = (2*4) + (3*1) = 8+3 = 11

Third part = 3*4 = 12

Carry process:

Right → 12 → write 2, carry 1

Middle → 11 + 1 = 12 → write 2, carry 1

Left → 2 + 1 = 3

Final answer → 322

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Example 2→

57 × 72

Here: d1=5, d2=7, d3=7, d4=2

```
      5 7
    X 7 2
-----------
35 | 10 + 49 | 14
```

First part = 5*7 = 35
Second part = (5*2) + (7*7) = 10+49 = 59
Third part = 7*2 = 14

Carry process:
Right → 14 → write 4, carry 1
Middle → 59 + 1 = 60 → write 0, carry 6
Left → 35 + 6 = 41

Final answer → 4104

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Example 3→

64 × 91

Here: d1=6, d2=4, d3=9, d4=1

```
      6 4
    X 9 1
  ---------
  54 | 6 + 36 | 4
```

First part = 6*9 = 54

Second part = (6*1) + (4*9) = 6+36 = 42

Third part = 4*1 = 4

Carry process:

Right → 4 → write 4

Middle → 42 → write 2, carry 4

Left → 54 + 4 = 58

Final answer → 5824

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Practice sheet

1. 36×24
2. 58×13
3. 47×62
4. 69×27
5. 84×15

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